The Copy Ceiling: An Input-Exposure Control for Ontology-Grounded Generation over Private Corpora

Dr John O'Hare
DreamLab AI

The Loom / VisionFlow neurosymbolic stack

August 17, 2026

Pre-print v2 (restructured). Supersedes v1; adds the negative-control arms and the production-node paired study.

Abstract

[ABSTRACT — written last. One-paragraph arc: private-corpus requirements; the instrument (copy ceiling, gain-over-copy); the uniformly negative ten-model result; the negative-control attribution; the OOD non-regression; the standard we propose.]

1 Introduction

A recurring enterprise requirement is a language model that answers accurately over a large, private, domain-specific corpus—a customer's product data, an organisation's internal knowledge, a curated scientific ontology—that the model cannot have memorised because it was never in pretraining. Retrieval-augmented generation is the standard remedy for this gap between parametric memory and the knowledge an answer requires [10, 21], and it helps most where parametric memory is weakest: long-tail and proprietary facts [24]. Structured, graph-based retrieval sharpens it further [4, 6, 13, 27]. But evaluating such a system faces a measurement problem the headline "grounding uplift" hides: when the gold answers are themselves derived from the corpus that is injected, a large raw-to-grounded gain conflates two very different behaviours—faithfully *delivering* the exposed facts, and *reasoning* over the injected structure. Published knowledge-graph grounding reports the gain without a standing control that separates them.

We introduce the *copy ceiling*—a per-item, graph-derived input-exposure baseline—and the signed *gain over copy*, the first control that separates faithful delivery of injected facts from reasoning over ontological structure in graph-grounded generation. For each question the copy ceiling is the recall a no-op extractor of *exactly* the text the model was shown would already score; the gain over copy is the model's grounded recall minus that ceiling. It is the retrieval-augmented, per-item, continuous generalisation of the discrete input-only baselines long used to interpret reading comprehension [18] and natural-language inference [14, 30]. Its rhetorical role is a control the reader calibrates the headline number against, and the claim we draw from it is methodological: grounding evaluations whose gold is derived from the injected corpus should report a copy ceiling as standard [25, 34].

The setting makes the control load-bearing. Unaided, the models we test score ≈ 0.26 on in-domain questions—the domain is genuinely private, absent from parametric memory—and ≈ 0.92 given the curated context. Almost the whole of that uplift is exposure: across ten contemporary models from five providers the gain over copy is small and *uniformly negative* (-0.067 to -0.022), so no model recovers gold beyond what a copy of its context already exposes, and they differ only in how much surfaced gold they drop. This is not a defect to explain away. High-fidelity delivery of an authoritative, human-vetted source *is* the product for private-knowledge grounding: the answer is trustworthy because the source is, attributable to a named generation, and cheap because the expensive curation is amortised once rather than re-paid per query.

Contributions.

1. **The instrument (C1, singular).** The per-item copy ceiling and the signed gain over copy: a formal definition, an explicit assumptions list, and an interpretation that reads delivery fidelity, lossy restatement and reasoning-over-structure off a single signed quantity (§5).

2. **The evidence (C2).** Ten models are uniformly negative on gain over copy; a four-arm negative-control design (true / shuffled / masked / irrelevant scaffolds) attributes the gain to scaffold *content*; and a production-node paired study isolates the ecosystem feature by holding the model constant and varying only the serving path (§6; results in §7).

3. **The organising frame (C3).** A conjunctive, multivariate bar—in-domain delivery fidelity *and* out-of-domain non-regression, measured on two different instruments—within which the copy ceiling is the in-domain axis (§2, §9).

4. **The artefacts (C4).** An open harness (paired bootstrap, Wilcoxon, Cliff's delta, Holm), the frozen question sets, all per-row data, and the served system itself.[1]

2 The Private-Corpus Setting

The design decisions this paper measures are not preferences; they are forced by a chain of deployment requirements. We state the chain so that the evaluation bar—and the system in §4—read as constraints rather than choices. Each requirement carries one design consequence and its architectural decision record (ADR); Table 1 records where each is enforced. The object of study is a served node[2] over a corpus built by a separate, CI-gated pipeline[3] and accelerated by an in-process HNSW index[4].

1. **Private-knowledge need ⇒ curated, amortised corpus.** The model must answer over a corpus it cannot have memorised (raw ≈ 0.26 is the proof of privacy); the answer is grounded in a human-directed, provenance-tracked ontology whose expensive curation is paid once and reused per query. The copy ceiling is the evaluation-time expression of "the curation, not the model, carries the answer" (ADR-135 D2; ADR-136 D4).

2. **Trust ⇒ human auditability at single-entity granularity.** A grounded answer is trustworthy only if a human can read and audit the served knowledge end-to-end; the canonical served unit is therefore one per-IRI markdown-with-ontology block—curated prose headed by its typed relations—and every retrieval port returns, or resolves to, an `Iri` addressing that unit (Invariant I-P1, "the prize"; RUST-ARCHITECTURE §0, ADR-136 D1).

3. **Auditability ⇒ accelerators strictly behind the markdown.** Nothing a machine indexes may become the thing a human trusts instead of the markdown; the lexical index, HNSW, oxigraph SPARQL and the attestation ledger are projections that find, rank or attest the unit and resolve back to its IRI, enforced by an acyclic crate ring rather than reviewer vigilance (ADR-136 D1–D3; RUST-ARCHITECTURE §0–1).

4. **Privacy / LAN-confinement ⇒ local delegation, no cloud on the hot path.** Content and answers must stay on the local network; generation is delegated to a LAN/local backend by URL and the augmentation hot path is fully in-process (ADR-135 D1; ADR-136 §3).

5. **Local-model churn ⇒ model swap without consumer change.** Local models change often (Gemma → Muse-Glimmer → Qwen3.8-27B); a stable OpenAI-compatible façade makes

[1] The Loom serving node: `https://github.com/DreamLab-AI/loom`.

[2] The served node `https://github.com/DreamLab-AI/loom` (code under the repository licence; corpus terms per the sibling knowledgeGraph repository).

[3] The canonical builder `https://github.com/jjohare/logseq` runs `pytest pipeline/tests` and `pipeline.validate` before publishing each generation.

[4] RuVector: `https://github.com/ruvnet/ruvector`.

Table 1: The private-corpus requirements arc: each business requirement, the design consequence it forces, and the point at which the consequence is enforced (not merely intended). ADR = the governing architectural decision record.

Requirement	Design consequence	Enforcement point
Private, unmemorised corpus	Human-curated ontology, curation amortised once	Sha-addressable generation; raw ≈ 0.26 (ADR-135 D2)
Human auditability	Per-IRI markdown-with-ontology as the served unit	Ports resolve to `Iri` → `CanonicalUnit` (I-P1)
Accelerators behind the unit	Indexes only find/rank/attest the markdown	Acyclic crate ring; adapter cannot mint the unit
LAN confinement	Model delegated by URL; in-process hot path	`DISTILL_BACKEND_URL`; Profile A network-free
Model churn	Stable façade; model identity in the result	OpenAI-compatible `/v1`; one config line
Bounded latency	No-model retrieval path	Lexical index 2.02 ms p50; scaffold needs no model
Injection interference	Confidence-gated selective injection	Single injection authority; skip below threshold
Never-mixed provenance	Per-answer attestation; ontology-only boundary	`loom` block + `corpusNature`; working graph never served

the backend one configuration line and lets model identity ride in the result, never the endpoint—which is exactly what lets this study hold grounding constant while varying the model under test (ADR-135 D1).

6. **Bounded latency ⇒ retrieval works with no model, fast.** Retrieval cannot depend on a model call: the scaffold, SPARQL and search endpoints need no model, and the lexical inverted index resolves in **2.02 ms** at the median over 8,146 class titles (ADR-136 §3).

7. **Injection interference ⇒ confidence-gated, benchmark-first injection.** Injected context can displace correct parametric knowledge on off-ontology queries [23, 33, 36]; a single injection authority scales the budget to the retrieval score and skips below a threshold [24], and features that could add noise are benchmark-gated (ADR-136 D3).

8. **Never-mixed provenance ⇒ attestation and boundary discipline.** Every grounded answer must trace to a named, versioned generation; each response carries a `loom` telemetry block and `corpusNature` honesty metadata, and the node serves only the published ontology, never the working graph (ADR-135 D6/D7; ADR-136 D5).

The bar these requirements imply is multivariate and conjunctive: excellent, attributable recall where the corpus is authoritative, *and* no regression on the general questions a model already answers. We introduce that framing here and reconcile the axes in §9.

3 Related Work

Knowledge-graph and private-corpus grounding. From the original RAG formulation [21] to graph-structured retrieval [6, 13], zero-shot knowledge-graph triple prompting [4], and the unifying roadmap and surveys [27, 29], the family grounds generation in retrieved graph content; GraphRAG motivates precisely our setting—answering over "private and/or previously unseen" collections [6]—and enterprise variants foreground the amortised-curation constraint [35]. Two features distinguish our object of study. Published work typically injects *instance* triples or opaque encodings—community summaries [6] or a GNN soft-prompt over a retrieved subgraph [15]—whereas we inject *schema-level* content (class definitions, typed relations, taxonomy) and keep the served unit a human-readable markdown block; and none of these reports an input-exposure control, so their uplift is uncontrolled for what the injected context already contains.

Baselines that reframe a result. A recurring genre installs a simple control and overturns the naive reading of a headline: a strong, simple RAG baseline matches elaborate pipelines under matched token budgets [34]; a controlled comparison reverses a prior "RAG wins" finding and adds a confidence router [22]; apparent argument comprehension is shown to be spurious-cue exploitation [26]. Our paper belongs to this genre: the copy ceiling is the control, and "uplift is exposure, not reasoning" is the reframing. The confidence router of Li et al. [22] and selective retrieval more broadly [3, 17, 24] are the design precedent for our injection gate—they route retrieve-versus-not; we gate inject-versus-abstain.

Input-only baselines and faithfulness metrics. The intellectual root of the copy ceiling is the input-only baseline: passage-only / question-only baselines reveal that much apparent competence is shortcut exploitation of what the input already exposes [18], as do hypothesis-only baselines in natural-language inference [14, 30]. Where these are discrete, per-task classifiers, the copy ceiling is per-item and continuous, for graph-grounded generation. On the metric side, faithfulness is separately measurable from correctness [1, 8, 38], and metric-first evaluations give the metric a formal definition with an explicit assumptions list [25] and a shortcut-robustness subsection [9]. FActScore's precision-against-a-source and Adlakha's token-overlap K-Precision are cousins of our exposure count; ALCE treats "copy the passage" as a shortcut to *reject*, whereas we make that copy the metric's reference point. Contamination audits [12, 31] measure how much of a score is visible in the input post hoc; we make the deliberate exposure the reference a priori.

The nearest neighbour, and how we differ. The closest prior work establishes, by controlled removal and injection of the gold answer, that answer presence drives retrieval-augmented gains [2]; a parallel line defines an accuracy-given-gold-context upper bound to filter leakage out of benchmarks [32]. These prove the *phenomenon* and use gold-context as a benchmark-cleaning device. We differ in the artefact and its use: the copy ceiling is a per-item recall of a no-op extractor of *exactly* the shown text, reported as a standing control rather than a cleaning step, and the signed gain over copy is read *across models* as a faithfulness axis on which raw uplift cannot separate them. We keep the exposure and measure against it; they remove it. Judged arms use a cross-family judge to avoid self-preference [28, 37].

4 The Ontology Scaffold

The system measured here is the shipped serving node: a single static Rust binary organised as an eight-crate hexagonal workspace—a pure `loom-domain` core holding the port traits, five adapters, a `loom-scaffold` policy crate and a thin `axum` façade. The crate ring is the enforcement mechanism for the audit invariant of §2: only the domain crate mints a `CanonicalUnit`, so an adapter physically cannot return its own row, triple or vector as the served payload (Invariant I-P1). The node was adversarially audited by a different model family (gpt-5.4) with all findings remediated, and the Rust port is pinned byte-for-byte to the retired Python original by golden fixtures.

The canonical unit and the corpus. The served, reviewable unit is one per-IRI markdown-with-ontology block: curated research prose headed by its typed relations (`subClassOf`, `requires`, `enables`, `uses`, `relatedTo`, `contrastsWith`). The corpus is one sha-addressable *generation*—**8,146 concept classes** and **282,492 triples** in the Whelk-reasoned closure at the generation evaluated here—mirrored atomically so the node always serves a known, mixed-build-free generation; it is not baked into the binary. We explicitly reject the trajectory in which knowledge degrades into opaque

Table 2: The request pipeline. Retrieval, gating and the scaffold serialisation need no model; only the final delegation calls one. The single injection authority (match + gate) is the sole path by which any candidate—lexical or, when enabled, semantic—reaches the model.

Stage	Mechanism	Model?
Match	Lexical inverted index over 8,146 class titles; 2.02 ms p50 → seed classes	no
Gate	Single injection authority: budget scaled to retrieval score, skip below threshold	no
Scaffold	Budget-clamped serialisation of the matched units' markdown-with-ontology	no
SPARQL / search	Read-only, prologue-aware LIMIT-clamped oxigraph over the reasoned closure	no
Semantic fallback	In-process HNSW over 384-d embeddings; fires on lexical miss → candidate seeds into the same gate	*default-off*
Delegate	Inject scaffold into the last user message; call `DISTILL_BACKEND_URL`	yes

community summaries or GNN-encoded subgraphs [6, 15]; the reviewable markdown is the moat, and the accelerators below only find, rank or attest it [11].

The lexical matcher and the confidence gate are the *sole* injection authority: every candidate flows through one policy that scales the scaffold budget to the retrieval score and injects nothing below a threshold—the mechanism by which the node avoids "going jagged" on off-ontology queries (§7.4). The graph store answers read-only, LIMIT-clamped SPARQL over the reasoned closure with no model call. An in-process HNSW semantic fallback over 384-dimensional embeddings is wired for out-of-vocabulary queries but ships *default-off*: its measured document-embedding recall (0.816) is below the 0.87 design floor, so it stays disabled pending a query-shaped embedding that clears a multivariate benchmark rather than a threshold relaxation—an honest red gate we return to in §10. Finally the model-swap seam: generation is delegated to whatever backend `DISTILL_BACKEND_URL` names (Qwen3.8-27B today), model identity rides in the response, and swapping the model changes no consumer—which is what lets §6 hold grounding constant and vary the model, or hold the model constant and vary the serving path.

5 The Copy Ceiling

The copy ceiling is the paper's primary contribution, so we give it a formal definition, an explicit assumptions list, and a robustness design of its own, following the metric-first precedent [9, 25].

5.1 Definition and assumptions

Fix a question q_i with gold target set G_i (the {slug, title} pairs the graph asserts as its answer), the exact scaffold text S_i shown to the model, and the model's answer a_i. Let $\mathrm{m}(t, X) \in \{0, 1\}$ be the deterministic surface matcher: $\mathrm{m}(t, X) = 1$ when the normalised title of gold item t, or at least 80% of its length-≥ 4 words, appears in text X. Define the exposed count and the per-item **copy ceiling**

$$e_i = \sum_{t \in G_i} \mathrm{m}(t, S_i), \qquad c_i = \frac{e_i}{|G_i|},$$

and the grounded recall $r_i = (\sum_{t \in G_i} \mathrm{m}(t, a_i))/|G_i|$. The copy ceiling is the recall of the no-op extractor that returns S_i verbatim as its answer: that answer scores exactly c_i by construction. For the raw arm S_i is empty, so $c_i \approx 0$.

Assumptions (when the instrument applies).

(A1) **Deliberate exposure.** Questions, gold and scaffold derive from one graph, so the gold is present in the injected text by design. The ceiling *measures* this exposure; it is not a leak to remove.

(A2) **Shared matcher.** c_i and r_i use the same matcher, so both are lower bounds under paraphrase, but their difference is first-order robust to the matcher's under-counting; where paraphrase is a live threat we re-score with a semantic judge (§6).

(A3) **Per-item, model-free ceiling.** c_i is a property of (q_i, S_i) only; for a fixed scaffold it is identical across models, which is what makes gain over copy comparable across a model sweep.

(A4) **Graph-derived gold.** The ceiling is informative only when gold is derived from the injected corpus. Where gold is authored independently of the scaffold (the out-of-domain arm), $c_i \approx 0$ and a different instrument—judged quality—carries the axis.

5.2 Gain over copy

The signed **gain over copy** is $g_i = r_i - c_i$, with headline $\bar{g} = \bar{r} - \bar{c}$. It turns delivery fidelity into a signed, model-discriminating quantity and reads directly:

- $g \approx 0$ — *delivery fidelity*: the model restates the exposed facts and fabricates almost nothing. For private-knowledge grounding this is the target behaviour, not a shortfall.
- $g < 0$ — *lossy restatement*: the model drops surfaced gold, a faithfulness failure; the magnitude ranks models where near-identical raw uplift cannot.
- $g > 0$ — would evidence *reasoning over structure*: recovering gold the exposure check missed, for example by composing a relation the scaffold implies but does not state. We observe none in-domain; it is the signal a multi-hop arm would target.

A near-ceiling result therefore licenses "faithful delivery" and forbids the inference "reasoning over ontological structure"—the distinction the rest of the paper turns on.

5.3 Negative controls: what would move the gain

A single objection can sink the instrument: that a positive-looking delivery is lexical overlap between answer and scaffold, not knowledge transfer. We answer it with four arms that hold the model and the question fixed and perturb only the injected block (defined here; results in §7.3):

true the block the node actually served, injected into the bare model. It should reproduce the live loom arm, proving the serving path adds nothing beyond the block itself.

shuffled the same block with its body sentences shuffled (wrapper kept, seeded). Same tokens, destroyed structure: if the gain survives, it was lexical soup.

masked the same block with every seed-class title replaced by `Entity-`k. Structure intact, entity names gone: if the gain survives, the score was matching names, not knowledge.

irrelevant the well-formed, on-corpus scaffold of a *different* question (a fixed derangement over the engaged items). If the gain survives, any ontology-shaped text would do.

The logic is differential. If gain over copy stays near zero under **shuffled**, **masked** and **irrelevant**, the ceiling is tracking exposure faithfully and the delivery is content-bound; if it moves, the ceiling was leaking through surface form or position. This is the experiment that turns the instrument from plausible to confirmed, and it is the direct analogue of a shortcut-robustness check for a faithfulness metric [9].

6 Method

Design. Both studies are paired within-model: every comparison is between two answers to the *same* question by the *same* model, so between-question difficulty variance cancels and the estimand is a paired per-question difference. The two studies isolate different variables—the ten-model sweep varies the *model* with grounding fixed; the new study varies the *serving path* with the model fixed.

Study 1 — the ten-model sweep (v1, summarised). Using the knowledge graph as its own oracle we template three question types from every sufficiently-defined class—**T-REL** (what a class requires / uses / depends on / enables; its parts), **T-TAX** (immediate and ancestral parents) and **T-COMMON** (a pair's shared ancestor)—with gold the {slug, title} targets the graph asserts. Seed 42 yields **510 questions** over 15 domains (285 T-REL, 180 T-TAX, 45 T-COMMON). Each is asked *raw* and *scaffold* (a 1,500-token budget-clamped ontology extract prepended), across ten models from five providers under one deterministic configuration (`temperature`=0, `max_tokens`=2048; `reasoning_effort`=low only where thinking cannot be disabled, so a mandatory pass does not truncate the answer). Scoring is the lexical matcher of §5: a gold item is a hit when its normalised title, or $\geq 80\%$ of its long words, appears in the answer. This under-counts paraphrase, so absolute recall is a lower bound and we treat the *paired* delta—same question, scorer and model—as the signal, flagged honestly as a lexical estimand.

Study 2 — the production-node paired study (new). To isolate the ecosystem feature rather than the model, we hold the model constant (Qwen3.8-27B on the reference GPU host) and vary only the serving path. The *loom* arm calls the production Rust node's `/v1/chat/completions`, which retrieves, confidence-gates and injects the scaffold and returns per-answer telemetry (engagement, injected tokens, fusion path) in the response `loom` block; the *raw* arm calls the same model backend directly with no scaffold. Both use `temperature`=0 and `max_tokens`=1536, below which the reasoning backend truncates to empty. Questions are three frozen sets—*arcane* (deep in-domain), *thin* (sparsely-covered) and *general* (out-of-domain)—of **117** questions in total (24 arcane, 33 thin, 60 general), reused from the v1 sweep so the arms are judge-comparable. The four negative-control arms of §5.3 run against the same node: scaffolds are fetched from the LLM-free `/loom/scaffold` endpoint—so every control injects exactly the block the loom arm served—and presented to the bare model as a system message.

Cross-family semantic judge. Because Study 2 spans in-domain and out-of-domain questions and must survive the paraphrase objection, answers are scored not lexically but by a reference-guided **0–5 rubric** judge. The judge is a different model family from the candidate—an OpenRouter GPT judge grading the Qwen candidate, never the candidate's own family—to avoid self-preference [28, 37]; the gold references are the graph-derived targets in-domain and frontier-authored key points out-of-domain.

Statistics. The headline per set and pooled is the paired mean difference with a seeded **10,000-resample bootstrap** 95% percentile interval [7, 19], complemented by a two-sided **Wilcoxon signed-rank** test and **Cliff's delta** as a distribution-free effect size. Because the three sets are reported together, set-level p-values carry a **Holm–Bonferroni** correction [5]. Where the claim is *equivalence* rather than difference—the out-of-domain non-regression axis—we use two one-sided tests (TOST) against a pre-registered margin [20] of $\pm 0.25, rather than reading anon - significantdifferenceasnoeffect.In-domain, graph-derivedquestionsclusterbyclassanddomain$ [5]; $followin$
$clusteredblockbootstrapalongsidethenaiveintervalandanintention-to-treatdeltathatretainsquestionsonwhic$

7 Results

7.1 In-domain: delivery fidelity

[TODO — cherry-pick v1 §Results in-domain]

7.2 Ten models: gain over copy

[TODO — cherry-pick v1 table + figure; add rank correlation]

7.3 Negative controls: attributing the gain

[TODO — NEW: true / shuffled / masked / irrelevant arms]

7.4 Out-of-domain: the gate holds

[TODO — cherry-pick v1 OOD + NEW general pairs]

8 Analysis: Exposure, not Reasoning

[TODO]

9 The Multivariate Bar

[TODO — shrunk scorecard]

10 Limitations and Threats to Validity

[TODO]

11 Conclusion

[TODO]

References

[1] V. Adlakha et al. "Evaluating Correctness and Faithfulness of Instruction-Following Models for Question Answering". In: *Transactions of the Association for Computational Linguistics* (2024). arXiv: 2307.16877. URL: https://arxiv.org/abs/2307.16877.

[2] *Answer Presence Drives Retrieval-Augmented Generation Rewriting Gains*. 2026. arXiv: 2606.05633. URL: https://arxiv.org/abs/2606.05633.

[3] A. Asai et al. "Self-RAG: Learning to Retrieve, Generate, and Critique through Self-Reflection". In: *ICLR*. arXiv:2310.11511. 2024.

[4] J. Baek, A. F. Aji, and A. Saffari. "Knowledge-Augmented Language Model Prompting for Zero-Shot Knowledge Graph Question Answering". In: *NLRSE Workshop, ACL*. 2023. arXiv: 2306.04136. URL: https://arxiv.org/abs/2306.04136.

[5] R. Dror, G. Baumer, S. Shlomov, and R. Reichart. "The Hitchhiker's Guide to Testing Statistical Significance in Natural Language Processing". In: *ACL*. 2018, pp. 1383–1392. DOI: 10.18653/v1/P18-1128. URL: https://aclanthology.org/P18-1128/.

[6] D. Edge et al. "From Local to Global: A Graph RAG Approach to Query-Focused Summarization". In: *arXiv preprint* (2024). arXiv: 2404.16130. URL: https://arxiv.org/abs/2404.16130.

[7] B. Efron. "Bootstrap Methods: Another Look at the Jackknife". In: *The Annals of Statistics* 7.1 (1979), pp. 1–26. DOI: 10.1214/aos/1176344552.

[8] S. Es, J. James, L. Espinosa-Anke, and S. Schockaert. "RAGAS: Automated Evaluation of Retrieval Augmented Generation". In: *EACL (System Demonstrations)*. 2024. arXiv: 2309.15217. URL: https://arxiv.org/abs/2309.15217.

[9] T. Gao, H. Yen, J. Yu, and D. Chen. "Enabling Large Language Models to Generate Text with Citations". In: *EMNLP*. arXiv:2305.14627. 2023, pp. 6465–6488.

[10] Y. Gao et al. "Retrieval-Augmented Generation for Large Language Models: A Survey". In: *arXiv preprint* (2023). arXiv: 2312.10997. URL: https://arxiv.org/abs/2312.10997.

[11] A. d'Avila Garcez and L. C. Lamb. "Neurosymbolic AI: The 3rd Wave". In: *Artificial Intelligence Review* 56.11 (2023), pp. 12387–12406. DOI: 10.1007/s10462-023-10448-w. URL: https://arxiv.org/abs/2012.05876.

[12] S. Golchin and M. Surdeanu. "Time Travel in LLMs: Tracing Data Contamination in Large Language Models". In: *ICLR*. 2024. arXiv: 2308.08493. URL: https://arxiv.org/abs/2308.08493.

[13] Z. Guo et al. "LightRAG: Simple and Fast Retrieval-Augmented Generation". In: *arXiv preprint* (2024). arXiv: 2410.05779. URL: https://arxiv.org/abs/2410.05779.

[14] S. Gururangan et al. "Annotation Artifacts in Natural Language Inference Data". In: *NAACL-HLT*. 2018. arXiv: 1803.02324. URL: https://arxiv.org/abs/1803.02324.

[15] X. He et al. "G-Retriever: Retrieval-Augmented Generation for Textual Graph Understanding and Question Answering". In: *NeurIPS*. 2024. arXiv: 2402.07630. URL: https://arxiv.org/abs/2402.07630.

[16] S. Hollis and F. Campbell. "What is Meant by Intention to Treat Analysis? Survey of Published Randomised Controlled Trials". In: *BMJ* 319.7211 (1999), pp. 670–674. DOI: 10.1136/bmj.319.7211.670.

[17] S. Jeong et al. "Adaptive-RAG: Learning to Adapt Retrieval-Augmented LLMs through Question Complexity". In: *NAACL*. arXiv:2403.14403. 2024.

[18] D. Kaushik and Z. C. Lipton. "How Much Reading Does Reading Comprehension Require? A Critical Investigation of Popular Benchmarks". In: *EMNLP*. 2018. arXiv: 1808.04926. URL: https://arxiv.org/abs/1808.04926.

[19] P. Koehn. "Statistical Significance Tests for Machine Translation Evaluation". In: *EMNLP*. 2004, pp. 388–395. URL: https://aclanthology.org/W04-3250/.

[20] D. Lakens. "Equivalence Tests: A Practical Primer for t Tests, Correlations, and Meta-Analyses". In: *Social Psychological and Personality Science* 8.4 (2017), pp. 355–362. DOI: 10.1177/1948550617697177.

[21] P. Lewis et al. "Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks". In: *NeurIPS*. 2020. arXiv: 2005.11401. URL: https://arxiv.org/abs/2005.11401.

[22] Z. Li et al. "Retrieval Augmented Generation or Long-Context LLMs? A Comprehensive Study and Hybrid Approach". In: *arXiv preprint arXiv:2407.16833* (2024).

[23] S. Longpre et al. "Entity-Based Knowledge Conflicts in Question Answering". In: *EMNLP*. 2021. arXiv: 2109.05052. URL: https://arxiv.org/abs/2109.05052.

[24] A. Mallen et al. "When Not to Trust Language Models: Investigating Effectiveness of Parametric and Non-Parametric Memories". In: *ACL*. 2023. arXiv: 2212.10511. URL: https://arxiv.org/abs/2212.10511.

[25] S. Min et al. "FActScore: Fine-grained Atomic Evaluation of Factual Precision in Long Form Text Generation". In: *EMNLP*. 2023. arXiv: 2305.14251. URL: https://arxiv.org/abs/2305.14251.

[26] T. Niven and H.-Y. Kao. "Probing Neural Network Comprehension of Natural Language Arguments". In: *ACL*. 2019. arXiv: 1907.07355. URL: https://arxiv.org/abs/1907.07355.

[27] S. Pan et al. “Unifying Large Language Models and Knowledge Graphs: A Roadmap”. In: *IEEE Transactions on Knowledge and Data Engineering* (2024). arXiv: 2306.08302. URL: https://arxiv.org/abs/2306.08302.

[28] A. Panickssery, S. R. Bowman, and S. Feng. “LLM Evaluators Recognize and Favor Their Own Generations”. In: *arXiv preprint arXiv:2404.13076* (2024).

[29] B. Peng et al. “Graph Retrieval-Augmented Generation: A Survey”. In: *arXiv preprint* (2024). arXiv: 2408.08921. URL: https://arxiv.org/abs/2408.08921.

[30] A. Poliak et al. “Hypothesis Only Baselines in Natural Language Inference”. In: **SEM*. 2018. arXiv: 1805.01042. URL: https://arxiv.org/abs/1805.01042.

[31] O. Sainz et al. “NLP Evaluation in Trouble: On the Need to Measure LLM Data Contamination for Each Benchmark”. In: *Findings of EMNLP*. 2023. arXiv: 2310.18018. URL: https://arxiv.org/abs/2310.18018.

[32] *SeedRG: Leakage-Free Benchmarks and a Gold-Context Upper Bound for Retrieval-Augmented Generation*. 2026. arXiv: 2605.08838. URL: https://arxiv.org/abs/2605.08838.

[33] F. Shi et al. “Large Language Models Can Be Easily Distracted by Irrelevant Context”. In: *ICML*. 2023. arXiv: 2302.00093. URL: https://arxiv.org/abs/2302.00093.

[34] *Stronger Baselines for Retrieval-Augmented Generation with Long-Context Language Models*. 2025. arXiv: 2506.03989. URL: https://arxiv.org/abs/2506.03989.

[35] *Towards Practical GraphRAG: Efficient Knowledge Graph Construction and Retrieval in Enterprise Environments*. 2025. arXiv: 2507.03226. URL: https://arxiv.org/abs/2507.03226.

[36] O. Yoran, T. Wolfson, O. Ram, and J. Berant. “Making Retrieval-Augmented Language Models Robust to Irrelevant Context”. In: *ICLR*. 2024. arXiv: 2310.01558. URL: https://arxiv.org/abs/2310.01558.

[37] L. Zheng, W.-L. Chiang, Y. Sheng, et al. “Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena”. In: *NeurIPS Datasets and Benchmarks*. arXiv:2306.05685. 2023.

[38] W. Zhou, S. Zhang, H. Poon, and M. Chen. “Context-faithful Prompting for Large Language Models”. In: *Findings of EMNLP*. 2023. arXiv: 2303.11315. URL: https://arxiv.org/abs/2303.11315.